

DCIT 208 — Software Engineering
Week 5 · Individual Assignment 2 (IA2)
Client Interview Evidence and Requirement Transformation Memo

Course Code:	DCIT 208 — Software Engineering
Assignment:	IA2 — Client Interview Evidence and Requirement Transformation Memo
Full Name:	Aidoo Robert Kwabena
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Team Name:	Techcare
Project Title:	Pharmacy Web Application
Assigned Feature:	Prescription Receipt Upload for Verification
Due Date:	Friday, 3 July 2026 — 5:00 PM Ghana Time
Date Submitted:	7 th July 3, 2026
Team Repo URL:	https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-23techcare
Individual Repo:	https://github.com/AJ2025LEG/CIT208-Client-Project-2026

AI Disclosure Statement

I used AI assistance to help structure and format this memo. The tool used was Claude (Anthropic). I used it to help organise my interview notes, structure the requirement transformation section, and draft the reflection based on my real conversation with a pharmacy professional. I reviewed, understood, and adapted all output to reflect my own project and personal understanding. I am able to explain and defend every part of this submission.

Date	Tool	Prompt Summary	Output Used	Rejected	Verification
7 th July 3, 2026	Claude (Anthropic)	Structure IA2 memo from real pharmacy interview notes	Section structure, requirement transformation, reflection draft	Generic examples not tied to my project	Read, understood and adapted to match my real interview content

SECTION A — Personal Question Set

The following 12 questions were prepared before conducting the interview with a pharmacy professional. Questions were designed to cover current workflow, pain points, data requirements, success criteria, security concerns, and failure handling — as required by the assignment brief.

Category: Current Workflow

1. Can you walk me through how a customer currently submits a prescription at your pharmacy?
2. How do you currently keep records of prescriptions you have received?

Category: Pain Points

3. Do customers ever bring fake or unclear prescriptions? How do you handle that?
4. What is the biggest problem with how prescriptions are handled right now?

Category: Data Requirements

5. What information on a prescription do you absolutely need to see before dispensing a drug?
6. What happens if a prescription photo is blurry or unreadable?

Category: Success Criteria

7. Would you be comfortable with customers uploading prescription photos online instead of bringing paper ones?
8. What would make you consider this prescription upload feature a success?

Category: Security & Privacy

9. Are there any privacy concerns about storing customers' prescription photos digitally?
10. Who should have access to uploaded prescription photos — only the pharmacist, or other staff as well?

Category: Failure & Exception Handling

11. How quickly do you expect to verify a prescription after it is uploaded?
12. What should happen if a prescription turns out to be expired or invalid after it is uploaded?

SECTION B — Interview Evidence

Date:	20 June 2026
Time:	3:00 pm
Medium:	In-person
Participant Role:	Licensed Pharmacist / Pharmacy Professional (name redacted for privacy)
Evidence Type:	User interview — direct questions and answer exchange
Location:	Pharmacy name/location redacted for privacy

Summary of Interview

I conducted an interview with a licensed pharmacy professional to gather firsthand information about how prescriptions are currently handled and what requirements an online prescription upload feature would need to meet. The conversation covered current workflow, common problems, data requirements, verification timelines, access control, privacy concerns, and success criteria. Key findings are summarized below.

Current Workflow

Customers currently bring paper prescriptions issued by a licensed doctor. A pharmacist checks the prescription for completeness and validity, verifies patient details and prescribed medications, and then prepares the medicine for dispensing. Some prescriptions are also received through email or messaging applications from healthcare providers.

Common Problems

The pharmacist identified four key challenges: customers frequently forget to bring their prescriptions; photos sent through messaging apps are often poor quality; significant time is spent verifying unclear prescriptions; and organizing and retrieving paper records is difficult.

Required Prescription Data

The pharmacist specified that the following information must be visible on every prescription before any medication is dispensed: patient full name, date of issue, drug name(s), dosage and strength, quantity prescribed, directions for use, doctor name, signature and license number, and hospital or clinic information where applicable.

Verification Timeline

The pharmacist expects prescriptions to be verified within 15 to 30 minutes during business hours, with a maximum of one hour during busy periods.

Access Control

The pharmacist stated that only the pharmacist should have full access to uploaded prescriptions. Authorised pharmacy assistants may access images only when necessary for processing orders, and all access must be controlled through user permissions.

Invalid Prescriptions

If a prescription is found to be expired or invalid, the customer must be notified immediately, the order must not proceed, and the system must record the reason for rejection.

Privacy and Security

The pharmacist confirmed that prescription data is confidential. The system must encrypt stored data, require secure login, restrict access to authorised staff, maintain audit logs of all record access, and comply with applicable healthcare data protection regulations.

Success Criteria

The feature would be considered successful if it reduces customer waiting time, makes verification faster, minimizes errors from unclear prescriptions, keeps records organised and easy to retrieve, protects customer privacy, and improves overall customer convenience and satisfaction.

SECTION C — Requirement Transformation

Raw Client Statement

"Yes, as long as the system is secure, the uploaded images are clear, and the pharmacist has the opportunity to verify the prescription before dispensing medication. Ideally within 15 to 30 minutes during business hours. During busy periods, verification may take up to an hour."

Functional Requirement

The system shall allow a registered customer to upload a prescription image (JPG or PNG, maximum 5MB). Upon upload, the system shall notify an authorised pharmacist to review and verify the prescription. The pharmacist shall be able to approve or reject the prescription from the admin dashboard. No medication order shall be processed until the prescription status is marked as Verified.

Non-Functional Requirement

The system shall send a prescription verification notification to the pharmacist within 2 minutes of upload. Prescription review and response shall be completed within 30 minutes during business hours (8am–6pm). The system should be available 99% of the time during business hours to ensure timely verification. All uploaded prescription images shall be encrypted at rest using AES-256 or equivalent standard.

User Story

As a pharmacy customer, I want to upload a photo of my prescription online so that I can order my medication without visiting the pharmacy in person and receive confirmation once a pharmacist has verified it.

Acceptance Criteria

AC1: Given a customer is logged in, when they navigate to the prescription upload page, then they can select and upload a JPG or PNG image not exceeding 5MB.

AC2: Given a prescription image has been uploaded, when the upload is successful, then the customer receives an on-screen confirmation message, and the pharmacist receives a notification within 2 minutes.

AC3: Given a pharmacist reviews an uploaded prescription, when the prescription is valid and complete, then the pharmacist can mark it as Verified and the customer's drug order is allowed to proceed.

AC4: Given a pharmacist reviews an uploaded prescription, when the prescription is expired, unclear, or invalid, then the pharmacist marks it as Rejected, the customer is notified with a reason, and the order does not proceed.

AC5: Given that an uploaded image is blurry or unreadable, when the system detects file quality issues or the pharmacist flags it, then the customer is prompted to upload a clearer image before the order can continue.

API / Interface Implication

The system requires a POST /upload-prescription endpoint that accepts multipart/form-data containing the customer's name, phone number, and image file. A separate PATCH /prescriptions/{id}/status endpoint is needed for the pharmacist to update the verification status (Pending, Verified, Rejected). The admin dashboard must display a queue of pending prescriptions with customer details and the uploaded image.

Data Implication

A prescriptions table is required in the database with the following fields: prescription_id (primary key), customer (foreign key), file path (path to stored image), upload timestamp, verification status (Pending / Verified / Rejected), rejection reason (nullable), verified by (pharmacist ID, nullable), and verified at (timestamp, nullable). Uploaded image files must be stored in a protected server directory, not publicly accessible without authentication.

Test Idea

Upload a valid, clear JPG prescription image as a logged-in customer and verify that: the file is saved correctly on the server; a new Pending record appears in the prescriptions table; the pharmacist dashboard displays the new prescription in the queue; and after the pharmacist marks it Verified, the customer's order status updates to allow progression. Additionally, test rejection by uploading a PDF or executable file to confirm the system rejects non-image formats with an appropriate error message.

Open Question

Should the system impose a business-hours restriction on prescription uploads — meaning uploads outside of 8am to 6pm are accepted but verification is only processed the next business day — or should the system display a clear message to customers about expected verification times at the point of upload? This needs to be confirmed with the client before implementation.

System-Design Mapping

Dimension	Implication
Interface	Upload form (customer-facing), prescription review queue (pharmacist dashboard), status notification UI
Data	prescriptions table with status tracking; encrypted file storage in protected server directory
Validation	File type (JPG/PNG only), file size (max 5MB), required fields (name, phone), login session check
Risk	Malicious file upload disguised as image; unauthorized access to prescription photos; pharmacist unavailable during business hours causing verification delays

SECTION D — Reflection

Before speaking with the pharmacist, I had a simple mental model of how the prescription upload feature would work. I assumed that a customer would take a photo of their prescription, upload it to the website, and the system would automatically check whether it was valid before allowing the drug order to proceed. I thought the verification process would be mostly handled by the system itself, with perhaps a basic check on the image file type and size. I also assumed that the pharmacist would only be involved in dispensing the medication, not in actively reviewing uploaded images through the application.

After the conversation, my understanding changed significantly. The most important shift was realising that prescription verification is a professional and legal responsibility that cannot be automated away. The pharmacist made it clear that a human must review every prescription uploaded before any medication is dispensed. This means the system needs to support a pharmacist-facing review queue — something I had not originally considered as a core part of the feature. What I thought was a simple upload form is a two-sided workflow: one interface for the customer to upload, and another for the pharmacist to verify.

One assumption I held before the conversation was that customers would mostly upload prescriptions from a laptop or desktop computer. This assumption became weaker after the interview. The pharmacist mentioned that customers often send prescription photos through messaging apps on their phones, which suggests that mobile accessibility is important. The upload interface must work well on small screens, and the system must handle images taken directly from a phone camera, which may vary widely in quality.

I also did not anticipate how detailed the data requirements would be. The pharmacist listed eight specific pieces of information that must be visible on every prescription — from the patient's name and drug dosage to the doctor's license number. This tells me that the verification status in the database cannot simply be a yes or no flag. There needs to be a way for the pharmacist to record why a prescription was rejected, so the customer can take corrective action. The open question that remains for me is whether the system should guide the pharmacist through a checklist of required fields during review or leave it entirely to their professional judgment. This is something I would want to confirm with the client in a follow-up conversation.
